



CSC1029 Object Oriented Programming

Lecture 2 – Procedural Programming Fundamentals (Part 1)





BEFORE WE BEGIN

A few things to mention ...

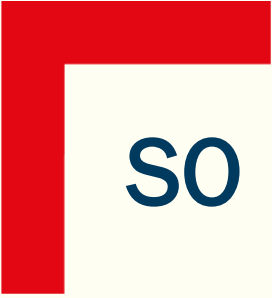
- a word about MS Teams chat
- do I really need to use Eclipse?
- quizzes
- we need to work on our communication 😊
- importance of lectures & practical sessions



WHAT'S TO COME TODAY?

A review of procedural programming concepts:

- program structure
- primitive data types
- non-primitive data types
- sequencing & selection



SO – WHAT'S THE PURPOSE OF THIS COURSE?

Hint – the focus is not to specifically turn you into a Java programmer.

While we do use **Java** as our development language ...

... together, we will practically examine the fundamental concepts of **OO** design and implementation.

The focus will be on **problem solving**.



WHAT IS A PROGRAM?

Input



Program

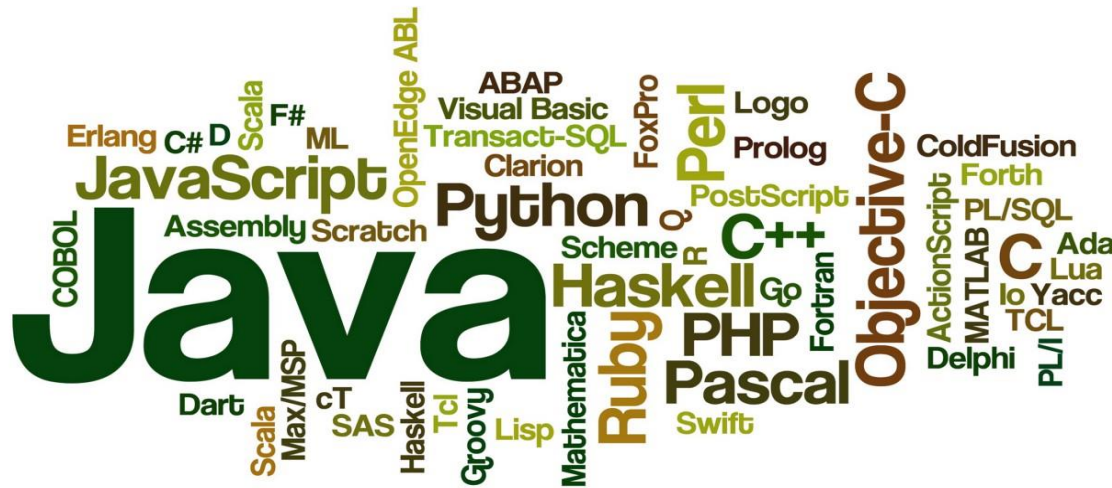


Output

YOU ARE A CPU – EXECUTE THE FOLLOWING PROGRAM!

1. Look at the person directly in front of you
2. Wait until they stand up
3. Now you stand up
4. Wait 2 seconds
5. Now sit down

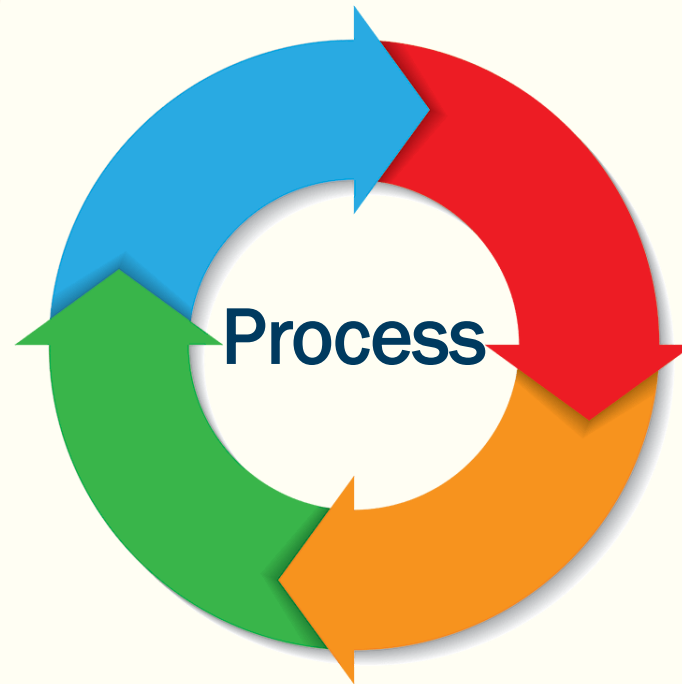
HOW DO WE MAKE A START? ⁽²⁾



Your main aim – for this course and future career – should be to think/design – **independently** of any specific language.

HOW DO WE MAKE A START? ⁽³⁾

Problem
Specification
↓
Design



I need a loop
in Java!

```
boolean finished = false;  
  
do {  
    finished = executeProcess();  
}  
while ( ! finished ) ;
```

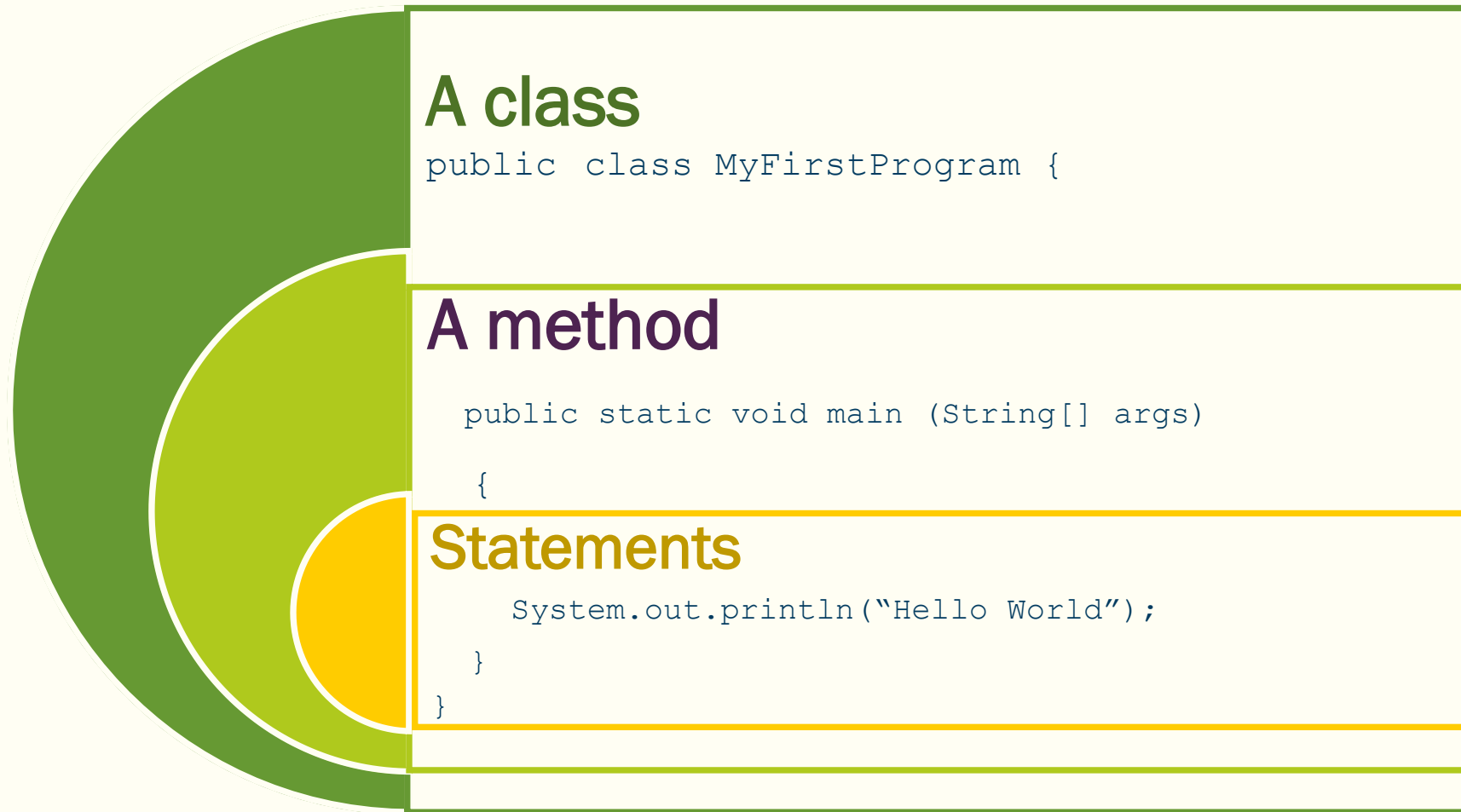
But what if I have
not done this
before?



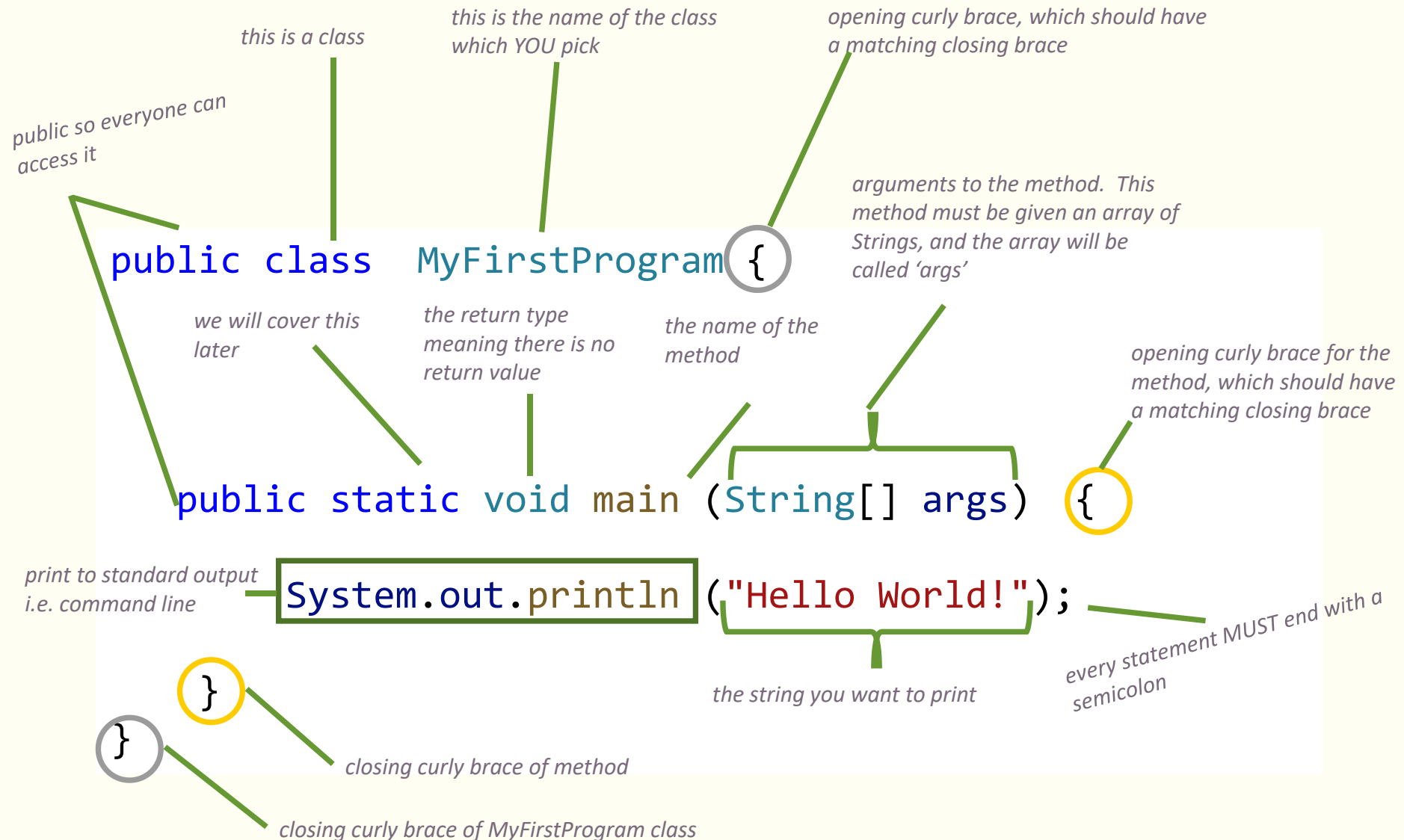
My First Program

```
public class MyFirstProgram {  
  
    public static void main(String[] args) {  
  
        System.out.println("Hello World!");  
  
    }  
}
```

Code Structure in Java



Anatomy of the source code



What allows the code to run?

- `main()`
- What does the `main()` method do?
 - *It instructs your computer to do something*

Do something	Do something again and again	Do something provided a condition
• statements • declarations • assignments • method calls	• Loops • For • While • Do	• If else • Switch • Nested Loops

- Is there a `main()` method in every class?
 - *No, only in the class you need to run the program*



How do we ask a user for information?

```
import java.util.Scanner;

Scanner in = new Scanner(System.in);

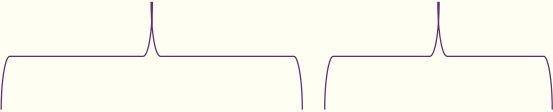
System.out.print("How much money are you lodging?");

int lodgement = in.nextInt();
```

Packages

import packageName.className;

Package Name Class Name



import java.util.Scanner;

import
statements must
be at the top of
the source file.

We have also created
our own packages and
this is a useful way to
organise your code.



Declaring variables: some rules ⁽¹⁾

- Variables **MUST** have a name
 - *We determine the name*
- Variables **MUST** have a type
 - *i.e. what is the content that we are saving in the variable? Is it a string, numbers, a true or false value?*
- Each type has a **size** associated
 - *This has an impact when saving the program*
- Variable types have **two** categories:
 - *Primitives*
 - *References to Objects*

Eight primitive data types in Java ⁽¹⁾

Four for different types of whole number:

<code>byte</code>	<code>aWeeNumber;</code>	<code>(-128 to +127)</code>
<code>short</code>	<code>aBiggerNumber;</code>	<code>(-32768 to 32767)</code>
<code>int</code>	<code>biggerAgain;</code>	<code>(-2147483648 to 2147483647)</code>
<code>long</code>	<code>theBiggest;</code>	<code>(-9223372036854775808 to 9223372036854775807)</code>



Eight primitive data types in Java ⁽²⁾

Two for different types of decimal number:

`float myAverage;` (7dp) -10^{38} to 10^{38}

`double myWage;` (15 dp) -10^{308} to 10^{308}



Eight primitive data types in Java ⁽³⁾

One for single characters:

char `oneCharacter;`

One for true or false values:

boolean `todayIsFriday;`





Declaring variables: some rules ⁽²⁾

- Variables must have a name
- Variables **MUST** have a type
 - *int*
 - *double*
 - *String*
 - *boolean*
- For example:
 - *int* *year*;
 - *double* *interest*;
 - *double* *balance*;

Examples ⁽¹⁾



PrimitiveTypes1.java

PrimitiveTypes2.java

PrimitiveTypes3.java

Non-Primitive Types : Strings

```
String name = "Michael"
```

- You can obtain the length of a string

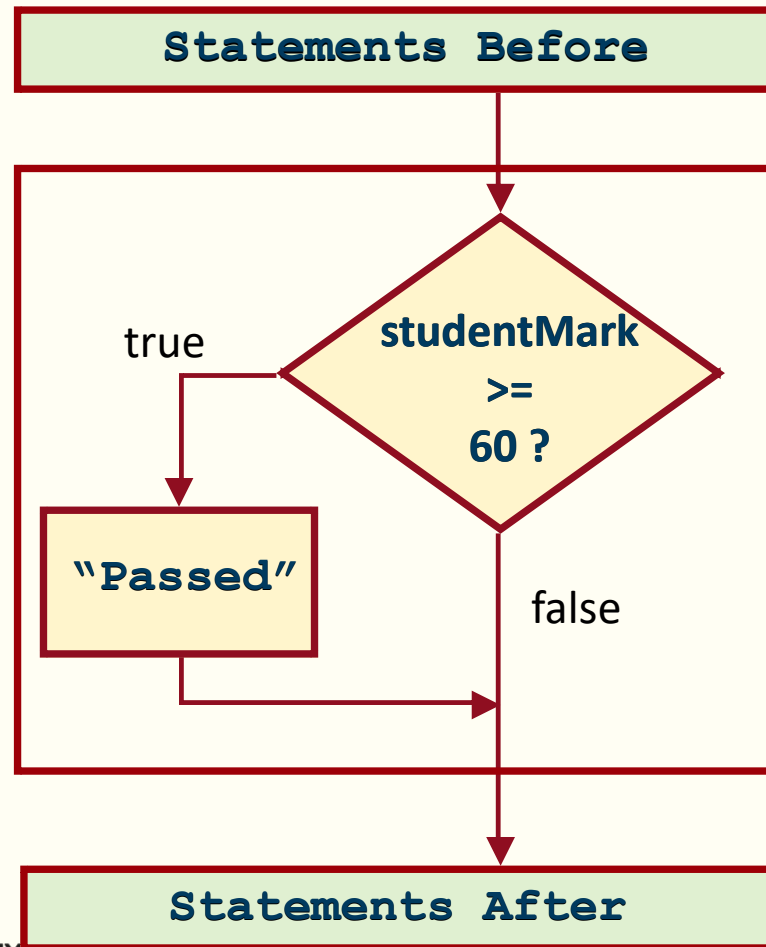
```
int n = name.length(); // will return 7
```

- A string of length 0 is an empty string and can be denoted by ""

- You can access a single character of a string

```
char letter = name.charAt(0);  
// will return 'M'
```

Program Structuring: Selection (if-statement)

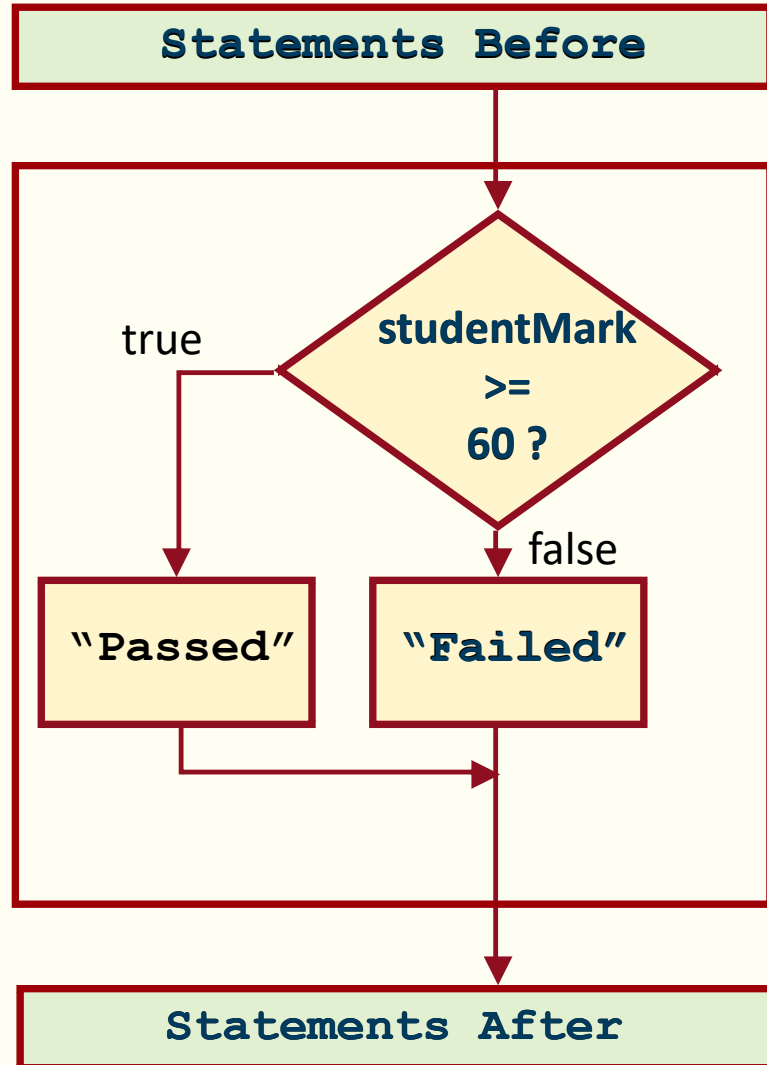


// Statements Before

```
if (studentGrade >= 60) {  
    System.out.println("Passed");  
}
```

// Statements After

Program Structuring: Selection (if/else-statement)



// Statements Before

```
if (studentGrade >= 60) {  
    System.out.println("Passed");  
}  
else {  
    System.out.println("Failed");  
}
```

// Statements After

Program Structuring: Selection (switch-statement)

```
switch (day) {  
    case 1:      dayOfWeek = "Monday";  
                break;  
    case 2:      dayOfWeek = "Tuesday";  
                break;  
    case 3:      dayOfWeek = "Wednesday";  
                break;  
    case 4:      dayOfWeek = "Thursday";  
                break;  
    case 5:      dayOfWeek = "Friday";  
                break;  
    case 6:      dayOfWeek = "Saturday";  
                break;  
    case 7:      dayOfWeek = "Sunday";  
                break;  
    default:     dayOfWeek = "Invalid";  
}  
// end switch
```

Each **case** we wish to consider is defined.

Each **case** is separated by the keyword **break**

Examples ⁽²⁾



Selection1.java

Selection2.java

Selection3.java



WHAT'S NEXT?

Iteration (Loops):

- while loops
- for loops
- do-while loops
- an example problem; 'WordWrap'